DAY 3- RAM AND STORAGE:

Tasks

1.

Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Typical speed	800–2133 MT/s	1600–3200 MT/s	4800 MT/s and above
Voltage	1.5 V	1.2 V	1.1 V
Bandwidth	Lower	Higher than DDR3	Much higher than DDR4
Power consumption	Higher	Lower than DDR3	Lower voltage and improved power management
Typical use	Older desktops and laptops	Mainstream PCs from the mid/late 2010s	Newer desktops, laptops and high-performa

Feature	DDR3	DDR4	DDR5
			nce systems
Compatibility	Older motherboards	DDR4-compatible motherboards	DDR5-compatible motherboards

2.
**Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.**

Safe procedure:

Shut down the computer, disconnect power, discharge static electricity, open the case, release the RAM clips, remove the module by its edges, inspect the notch, align it with the slot and press evenly until the clips lock. Reconnect power and boot the system.
Confirm the installed memory in the BIOS/UEFI or operating system.

3.

List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	HDD	SSD	NVMe SSD
1. Technology	Uses spinning magnetic disks and a moving read/write head	Uses flash memory with no moving parts	Uses flash memory and communicates through PCIe/NVMe
2. Speed	Slowest	Faster than HDD	Usually the fastest of the three
3. Cost	Cheapest per GB	More expensive than HDD	Usually more expensive than HDD and can cost more than SATA SSD
Daily use / app	Google Drive/local	Spotify, Windows and	Games like GTA V or large

Feature	HDD	SSD	NVMe SSD
	backup storage	everyday applications	software such as Android Studio

4.
**Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.**

Step 1: Shut down the computer

- Save your work and completely shut down the computer.
- Disconnect the charger/power cable.
- Remove any USB devices and other peripherals.
- If it is a desktop, switch off the PSU before unplugging it.

Step 2: Prevent static electricity

- Work on a clean, dry table.
- Touch a grounded metal surface or use an anti-static wrist strap.
- Avoid working on carpets.
- Handle the storage drive only by its edges.

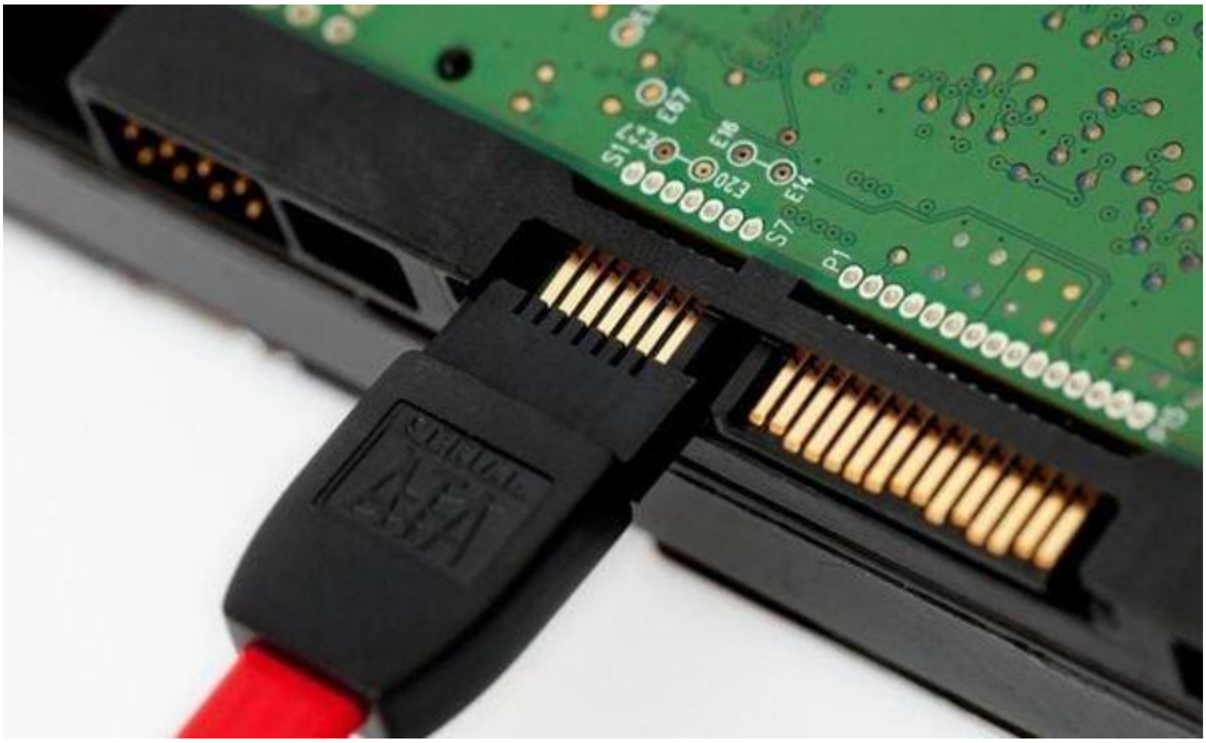
Step 3: Open the computer

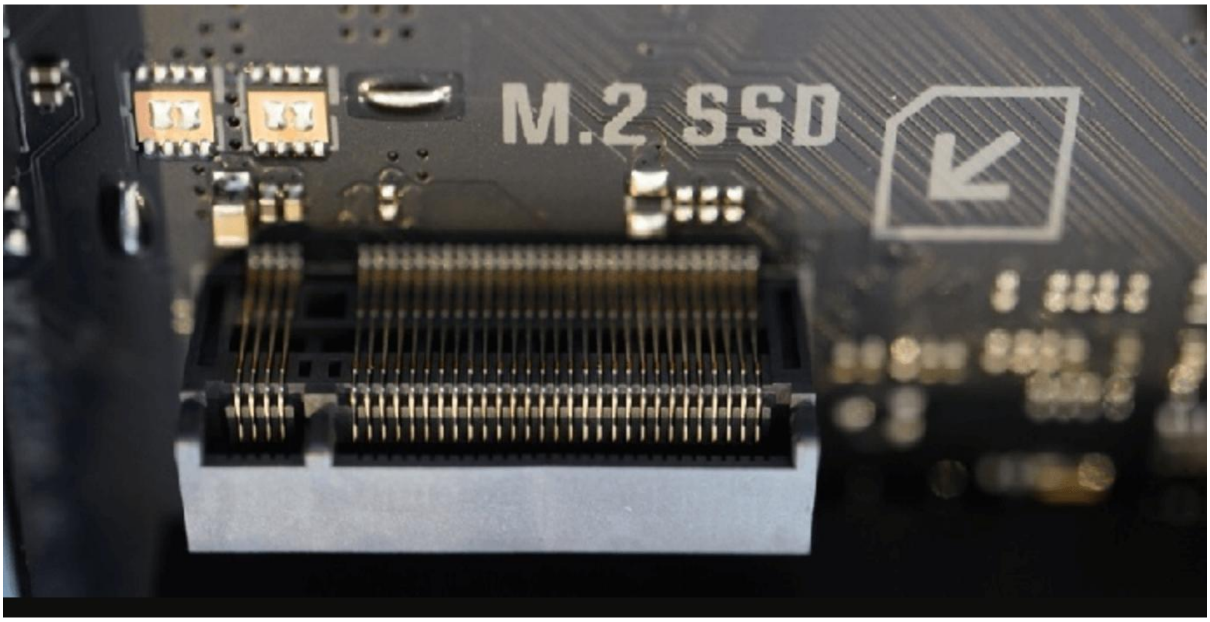
- Remove the appropriate screws from the laptop bottom cover or desktop side panel.
- Carefully remove the cover.
- Do not force the cover if it doesn't come off.

Step 4: Locate the storage drive

There are two common types:

Type	**What it looks like**
SATA HDD/SSD	Usually a rectangular drive connected using a SATA data connector and SATA power connector
M.2/NVMe SSD	Small stick-shaped circuit board screwed directly onto the motherboard





6

Step 5: Take the required photo 📸

Before removing anything:

1. Take a clear photo showing the storage drive.
2. Take a close-up photo of the connector.

3. Identify it as:
    - SATA
    - M.2
    - NVMe

   You can write something like:
   "The installed storage drive is an M.2 NVMe SSD. It is connected directly to the motherboard through an M.2 slot and is secured with a small screw."

Step 6: Remove the drive

For a SATA drive:

- Disconnect the SATA data cable.
- Disconnect the SATA power cable.
- Remove the mounting screws/bracket.
- Carefully remove the drive.

For an M.2/NVMe SSD:

- Remove the small retaining screw.
- The SSD will lift slightly.
- Pull it out gently at an angle.

Step 7: Reinstall the drive

- For SATA: reconnect the SATA data and power cables and secure the drive.
- For M.2/NVMe: insert the SSD into the slot at an angle, press it down gently and reinstall the retaining screw.
- Make sure the connector is fully seated.

Step 8: Close the computer

- Check that no cables are trapped.
- Replace the cover.
- Tighten the screws.
- Reconnect the power/charger.

Step 9: Confirm the storage is recognized

Turn the computer on.

On Windows:

1. Press Win + X.

2. Select Disk Management.
3. Check whether the storage drive appears.
   You can also check:
   Settings → System → Storage
   or
   File Explorer → This PC
   If the drive appears and your files are accessible, the reinstallation was successful.